Rithish S

SDE Intern - Backend | Scalable APIs, Distributed Systems, Node.js & Databases

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Education

Bannari Amman Institute of Technology — Sathyamangalam, Tamil Nadu
Bachelor of Engineering in Computer Science and Engineering — *Aug. 2024 – May 2028*
CGPA: **8.28/10**

Technical Skills

Languages & Core CS: JavaScript (ES6+), TypeScript, Node.js, Go, Python, C++, Java, SQL, OOP, OS, Networks
Backend & Systems: Fastify, Express.js, REST APIs, Microservices, Clean Architecture, Zod Validation, JWT Auth
Databases & Distributed Systems: PostgreSQL, MySQL, MongoDB, SQLite, Redis (Distributed Queues & Locks), BullMQ, Prisma
Cloud, DevOps & Security: Docker, Kubernetes, AWS (S3, EC2), GitHub Actions, Linux (POSIX), AES-256-GCM, Git
Certifications: AWS Cloud Practitioner Essentials, Oracle Cloud Foundations Associate, Cisco Linux Unhatched

Experience

Web Developer — Feb. 2025 – Dec. 2025
Students Special Group — *Student Organization*

- Designed and implemented backend REST APIs and normalized relational schemas for a Faculty Dashboard with MySQL, optimizing query throughput and relational consistency.
- Engineered full-stack internal management portals with React, Node.js, and MySQL, ensuring high availability and automated workflows for academic operations.
- Mentored **50+ students** in backend architecture, Node.js REST API patterns, database modeling, and software engineering best practices.

Projects

DB Backup CLI (DBVault) | *TypeScript, Node.js, Redis, BullMQ, AWS S3, Docker* GitHub

- Architected an automated multi-database disaster-recovery utility published on npm and Docker Hub with **120 passing automated tests** across PostgreSQL, MySQL, MongoDB, and SQLite.
- Designed a distributed job-processing system using Redis and BullMQ with configurable worker concurrency and queue limits to process backup workloads reliably.
- Implemented **AES-256-GCM encryption**, local key rotation, SHA-256 checksums, and Redis-based distributed locking; integrated cloud storage adapters for automated offloading to **AWS S3**.

PatentIQ | AI Prior-Art & Novelty Engine | *Node.js, Fastify, PostgreSQL, Prisma, Pinecone, Ollama* GitHub

- Engineered a high-performance backend microservice with Fastify, Prisma ORM, and PostgreSQL utilizing Clean Architecture, dependency injection, and connection pooling.
- Built a semantic search and RAG analysis pipeline integrating Pinecone vector database and local LLMs (Qwen2.5) to generate structured, deterministic technical risk assessments.
- Implemented secure REST endpoints with Zod schema validation, JWT-based authentication, and structured error handling for high-throughput query processing.

Open Source

ESLint | *JavaScript, AST, Static Analysis, IEEE 754, CI/CD* PR #21218

- Contributed a merged fix to ESLint's `no-loss-of-precision` rule, distinguishing genuine zero literals from non-zero decimals that underflow under IEEE 754 binary64 specifications.
- Added a comprehensive regression test suite covering underflow, subnormal values, signed zero, and non-decimal literals; passed **43 CI checks** before merge into a world-class codebase.

Achievements

- **1200+ DSA Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, spanning arrays, dynamic programming, trees, and graphs.
- **IEEE DevSpark Hackathon Winner** – Secured **1st place among 120+ teams** in Full Stack Development by building Plantera, an AI-powered environmental monitoring application.
- **BIT Hackathon Winner** – Secured **1st place among 200+ teams** by architecting a full-stack Leave Management System under hackathon time constraints.
- **Official Cloud Certifications** – Certified in **AWS Cloud Practitioner Essentials** and **Oracle Cloud Infrastructure Foundations Associate** (2025).